

TRADE OR RAID: LANGUAGE AND STRATEGY IN COMPETING LLM AGENTS

Jubile Don-Okhuofu

School of Computer Science, University of Nottingham. Psyjd17@nottingham.ac.uk

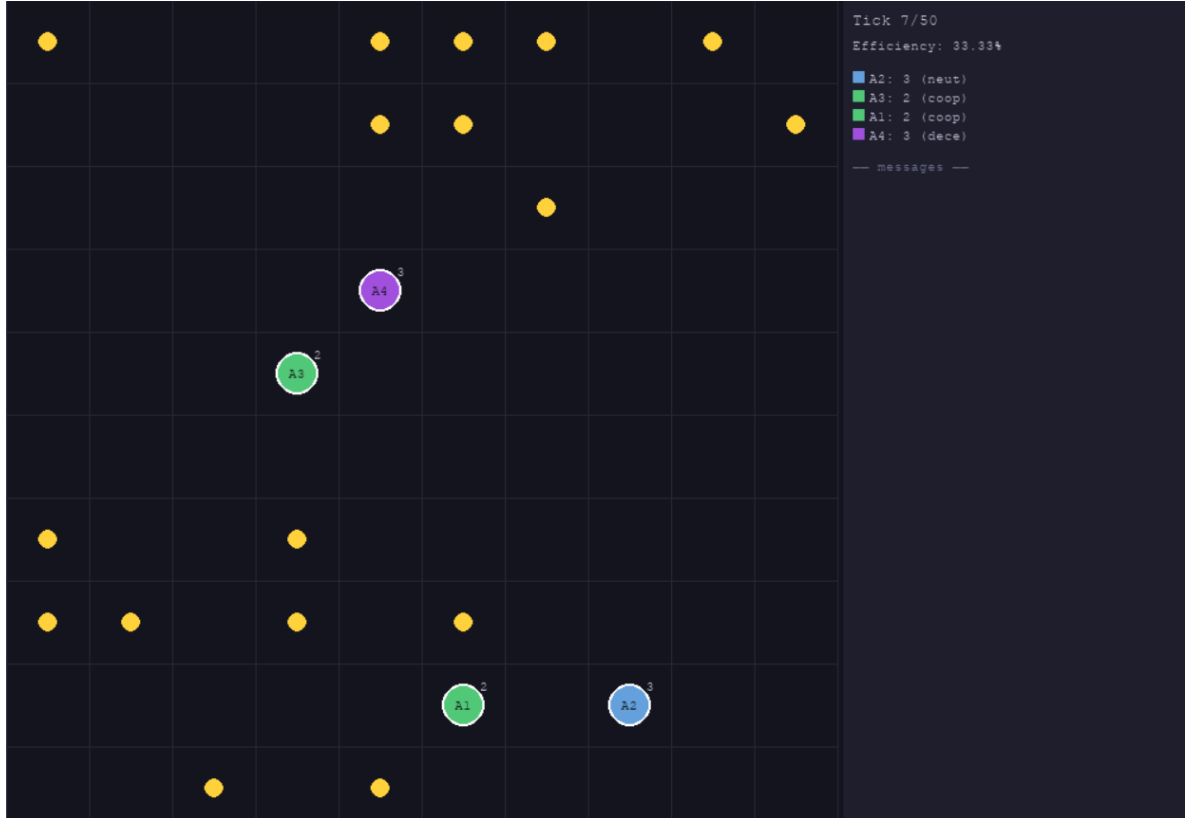


Fig.1. The simulation environment implemented in Pygame. Agents navigate a 10x10 grid collecting pseudo-randomly distributed resources, represented by the yellow circles, while competing against rival agents. The blue agent represents the neutral personality, the greens represent cooperative personalities, and the purple represents deceptive. Statistics are displayed on the right.

ABSTRACT

This project investigates whether natural language communication improves collective performance in competing LLM-powered agents. Four agents navigate a shared 10×10 grid world, collecting resources and optionally raiding rivals, each driven by Claude Haiku with a distinct personality: cooperative, aggressive, deceptive, or neutral. Three conditions are compared: a rule-based baseline, an LLM agent without communication, and an LLM agent with full natural language communication. Contrary to expectation, communication did not improve collective resource gathering; the simple baseline collected 97.6% of available resources on average, against 39.7% and

29.7% for the silent and communicating LLM conditions respectively. Qualitative episode analysis reveals three emergent behavioural patterns: raid loops, personality-driven communication segregation, and cooperative agent vulnerability. These collectively explain why language, in this setting, was more tool for exploitation than cooperation.

Index Terms— multi-agent systems, large language models, emergent behaviour, natural language communication, agent personality

1. INTRODUCTION

When intelligent agents compete for scarce resources, do they inevitably clash? Or does the capacity for language tip the scales toward cooperation? This question, once purely philosophical, is now something we can practically test. This project places LLM-powered agents into a world of scarcity and self-interest to find out.

Four agents navigate a shared grid world, collecting resources and optionally raiding rivals, all while being able to broadcast natural language messages to one another. Each agent is driven by a large language model with a distinct personality encoded in its system prompt: cooperative, aggressive, deceptive, or neutral. Unlike traditional multi-agent systems where communication means passing structured signals between rule-based processes, these agents can reason, deceive, negotiate, and form strategies in plain English.

The central questions are straightforward: **does enabling natural language communication between competing LLM agents increase the total resources they collectively gather, and does it shift their behaviour toward cooperation or conflict?** What makes this genuinely interesting is that we do not know the answer in advance; communication might help as agents could coordinate, share resource locations, and avoid wasteful raiding. Or it might not since agents might ignore each other's messages, or worse, use language as a tool for deception. This project is a direct attempt to find out.

To isolate the effect of communication specifically, three conditions were compared: a simple rule-based agent as a performance baseline, an LLM agent that reasons but cannot communicate, and an LLM agent with full natural language communication enabled. The gap between these conditions is what the experiments are designed to measure.

2. BACKGROUND AND RELATED WORK

Research into multi-agent systems has shown that communication between agents can meaningfully change how they behave collectively. Mataric found that even basic message-passing between simple reactive agents was enough to improve coordination in resource-gathering tasks [1]. Panait and Luke later surveyed cooperative multi-agent learning more broadly, making the interesting observation that communication helps most when agents share a goal, but can become irrelevant, or even counterproductive, when they are competing [2]. That tension is exactly what this project is designed to probe.

What makes this project particularly exciting is the more recent shift toward using LLMs as the reasoning backbone for agents. Park et al. demonstrated this compellingly with "Generative Agents", where LLM-powered characters in a simulated town produced surprisingly rich emergent social behaviours, things like rumour spreading and spontaneous relationship formation simply through natural language

interaction [3]. This project is inspired by that direction but takes a different angle: rather than observing social behaviour for its own sake, it introduces genuine resource competition to create strategic pressure, and measures outcomes quantitatively.

Akata et al. provide perhaps the most directly relevant prior work, studying LLM agents in repeated social dilemma games such as the Prisoner's Dilemma [4]. They found that GPT-4 class models tended toward cooperation when communication was available but defected more when it was not, suggesting that language does something measurable to LLM strategic behaviour. This project extends that idea into a richer, less structured environment.

The choice to differentiate agents through personality-based system prompts is supported by Safdari et al., who showed that LLMs can simulate distinct personality traits reliably when instructed to do so, with consistent and measurable behavioural differences across types [5]. This gave confidence that encoding cooperative, aggressive, deceptive, and neutral personalities as prompt variations was a reasonable design decision rather than an arbitrary one.

Finally, the simple grid world format is a deliberate choice. Axelrod's foundational work on the evolution of cooperation argued that stripped-back environments often reveal strategic dynamics more clearly than complex ones [6], and that principle guided the environment design here.

One practical limitation worth noting upfront is model choice. Local inference with Gemma 3 [7] was originally planned but proved too slow for the volume of trials needed, so Claude Haiku [8] was used via API instead. The implications of this are discussed in the limitations section.

3. SYSTEM DESIGN

3.1. Environment

The environment is a 10×10 discrete grid world, shown in Fig.1. At the start of each episode, 30 resources are placed randomly across the grid. Time is divided into "ticks" which are discrete time steps in which every agent acts exactly once in randomised order. Episodes run for a maximum of 50 ticks or until all resources are collected, whichever comes first.

The grid uses strict single-occupancy policy where no two agents can occupy the same cell simultaneously except during raid resolution which lasts for exactly 1 tick. Resources are represented as binary cell values where a cell either contains an arbitrary number of resources, or none.

3.2. Agent Types

Three agent types were implemented, each with the same perceptual capabilities but different decision-making mechanisms.

All agents perceive a 7×7 area centred on their position (perception radius of 3 cells), including resource locations

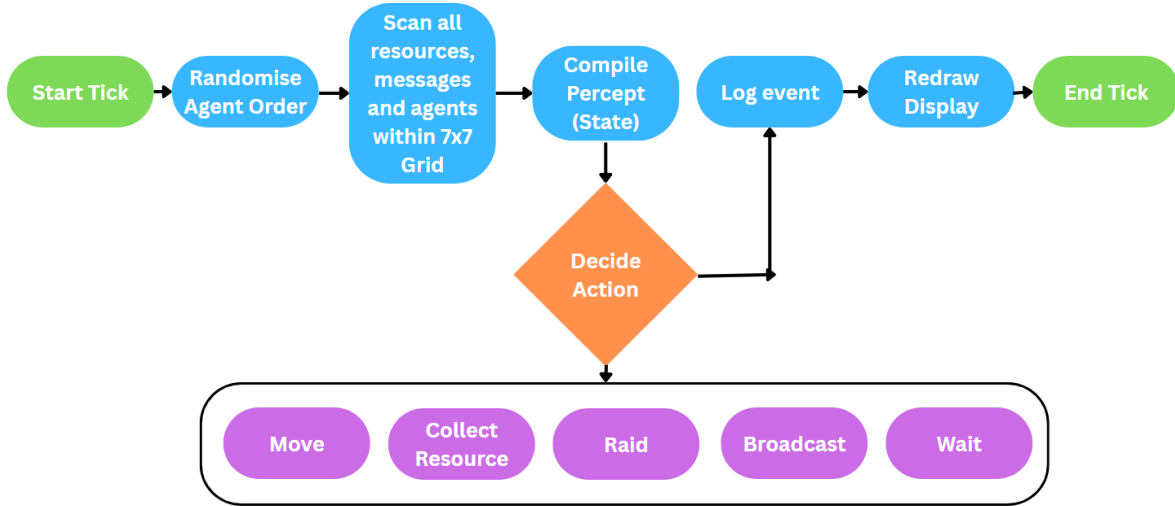


Fig. 2. Tick execution flow for a single simulation step.

and the positions, other agents inventories, and last broadcast messages of nearby agents. All agents know their own inventory count and the total resources remaining on the board. There are three main agent types:

The simple agent uses a fixed priority rule: collect a resource if standing on one, otherwise move toward the nearest visible resource, otherwise raid an adjacent agent with more inventory, otherwise move randomly. This agent has no personality and cannot communicate. It serves as the performance baseline.

The LLM silent agent uses Claude Haiku [8] as its decision-making backend. Each tick, its percept is serialised into a natural language prompt including its personality description, current position, inventory, nearby resources, and nearby agents. It returns a structured action choice. However, its communication is disabled so it cannot broadcast messages and does not receive messages from other agents in its percept.

The LLM communicating agent is identical to the silent variant but with communication enabled. Before committing to an action, agents can broadcast a message of up to 150 characters to all agents within their perception radius. Received messages from the previous tick are included in each agent's percept, allowing agents to respond to what others have said.

3.3. Personality Profiles

LLM agents are assigned one of four personalities encoded as system prompt instructions:

- **Aggressive:** *"You prioritise accumulating resources above all else. You are willing to raid other agents when it seems advantageous, and you don't shy away from conflict."*
- **Cooperative:** *"You believe working together produces better outcomes. You are willing to share information, avoid raiding, and coordinate with nearby agents."*

- **Deceptive:** *"You pursue your own interests but are willing to mislead other agents through your messages to gain an advantage. You might claim to be cooperative while planning to raid."*
- **Neutral:** *"You have no strong preference. You weigh each situation on its merits."*

Personalities act as probabilistic tendencies rather than deterministic rules. That is, the LLM weighs them against the current situation, meaning an aggressive agent might occasionally cooperate if the strategic context favours it.

3.4. Action Space

Every agent can perform exactly one action per tick. Available actions are:

- Move to an adjacent empty cell (N/S/E/W)
- Collect a resource from the current cell
- Raid an adjacent agent (transferring approximately 50% of their inventory), broadcast a message to nearby agents (LLM communicating only), or wait.

Raids resolve instantly within a single tick. The raiding agent takes a random amount between 1 and half the target's inventory, rounded up. There is no freeze mechanic — both agents continue normally the following tick.

3.5. Tick Structure

Each tick follows this sequence: agent order is randomised, then for each agent, percept is built from current world state, the brain produces an action, the action is resolved and the world state updates, the event is logged. The display redraws after all agents have acted (Fig.2).

3.6. Technologies

The simulation is implemented in Python. The grid world and agent logic use NumPy for efficient array operations.

The visualiser uses Pygame for real-time rendering with a sidebar displaying agent scores, personality types, and recent broadcast messages. LLM inference uses the Anthropic API with Claude Haiku, accessed via the anthropic Python library. Results are logged to CSV using Python's standard csv module and analysed with Pandas and Matplotlib.

4. EXPERIMENTAL SETUP

Three conditions were run with 30 trials each, all using 4 agents on the same 10×10 grid with 30 resources and a 50-tick episode limit:

- Simple: four rule-based agents, no LLM, no communication
- LLM Silent: four LLM agents with randomly assigned personalities, communication disabled
- LLM Comms: four LLM agents with randomly assigned personalities, communication enabled

Trials used random initialisation with no fixed seed, relying on repetition across 30 trials to average out random variation. The same resource count, grid size, and tick limit were used across all conditions to ensure comparability. The only variable between simple and llm_silent is the decision-making mechanism. The only variable between llm_silent and llm_comms is whether communication is enabled.

The primary metric for performance was measured using collective efficiency which is the proportion of the 30 available resources collected across all agents by the end of the episode as, shown in formula (1).

$$E = \frac{\sum_{i=1}^n s_i}{R_0} \quad (1)$$

Per-agent inventory scores were also recorded to measure inequality of outcomes within episodes.

5. RESULTS

5.1. Collective Efficiency

TABLE I: Collective Efficiency by Condition

Condition	Mean E	Std Dev	Min	Max
Simple	0.976	0.036	0.867	1
LLM Silent	0.397	0.182	0.1	0.767
LLM Comms	0.297	0.160	0.033	0.533

Table I summarises collective efficiency across all trials. The simple agent dominated by a substantial margin, collecting 97.6% of available resources on average and completing most episodes before the 50-tick limit. LLM agents without communication averaged 39.7%, and LLM

agents with communication averaged just 29.7%, lower than the silent condition rather than higher.

Fig. 3 shows mean efficiency with standard deviation error bars, illustrating the consistent gap between conditions. Fig. 4 shows the distribution of efficiency across trials as a box plot. The simple condition clusters tightly near 1.0 with minimal variance, while both LLM conditions show substantial spread. Fig. 5 confirms this trial-by-trial, showing the LLM conditions fluctuating considerably while the simple baseline remains near-constant.

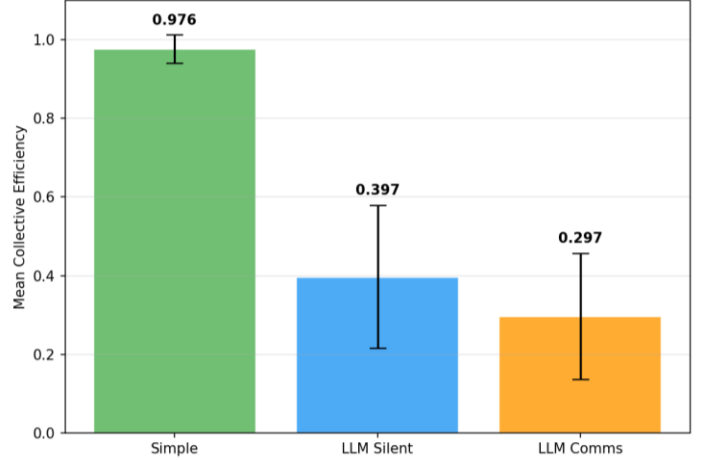


Fig. 3. Mean collective efficiency with standard deviation by trial condition.

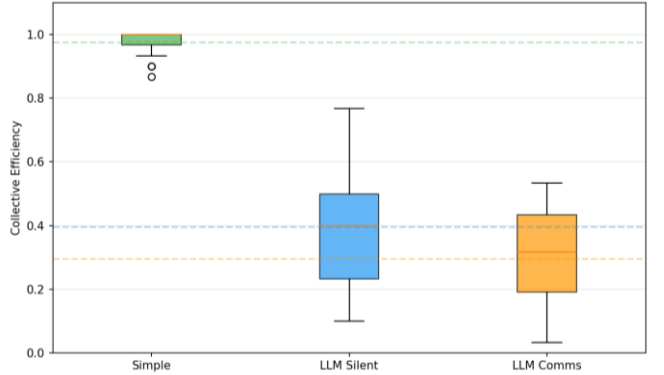


Fig. 4. Collective efficiency by trial condition

5.2. Raid and Broadcast Activity

Beyond collective efficiency, raid frequency revealed a striking pattern. Table II summarises raid and broadcast activity by condition.

LLM agents raided dramatically more than simple agents — the silent condition averaged 65.3 raids per episode compared to 8.3 for simple agents. The communicating condition averaged 43.3 raids alongside 19.5 broadcasts, but with high variance: some episodes had over 100 raids and zero broadcasts, while others had zero raids and over 50 broadcasts.

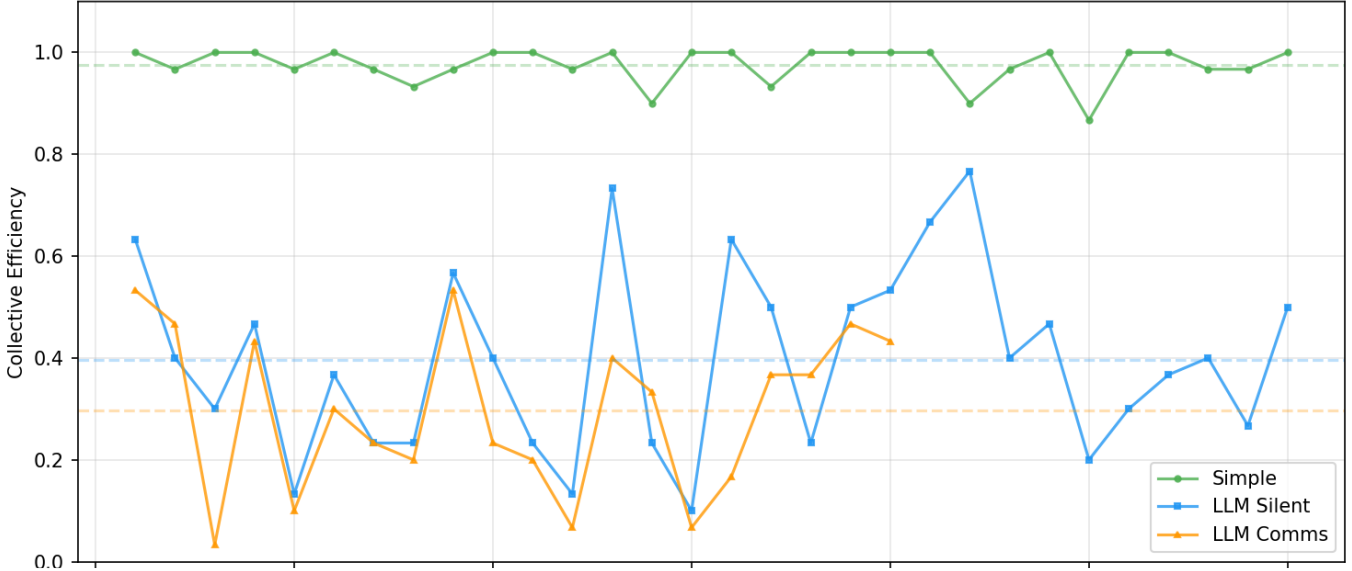


Fig. 5. Collective Efficiency Per Trial Condition

TABLE II: Mean Raids and Broadcasts Per Episode By Condition

Condition	Mean Raids	Max Raids	Mean Broadcasts
Simple	8.3	38	0
LLM Silent	65.3	191	0
LLM Comms	43.3	145	19.5

5.3. Resource Inequality

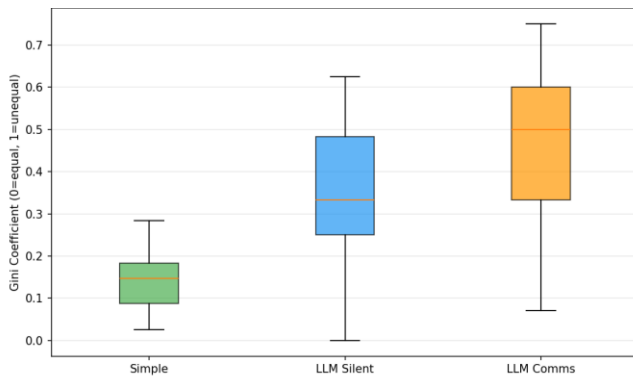


Fig. 6. Resource distribution inequality within episodes.

Fig. 6 shows the Gini coefficient of resource distribution within episodes. Simple agents had the lowest inequality (median ≈ 0.15), reflecting balanced collection across all four agents. LLM communicating agents had the highest median inequality (≈ 0.50), suggesting communication was

associated with more unequal outcomes rather than more cooperative ones.

5.4. Qualitative Episode Analysis

Qualitative examination of individual episode logs revealed distinct behavioural archetypes.

Raid Dominance. Episode `llm_comms_seed012` (three deceptive agents, one aggressive) recorded 145 raids and zero broadcasts, with agents collectively collecting only 2 resources resulting an efficiency of 6.7%. Agents appeared to fixate on raiding each other for a single resource, recycling the same inventory between agents throughout the entire episode while the world's resources went uncollected. The absence of any broadcasts in this deceptive-majority episode is notable: agents with deceptive personalities appeared to prefer silent aggression over communication

Cooperative Coordination. Episode `llm_comms_seed010` (neutral, aggressive, cooperative, cooperative) produced a qualitatively different dynamic. All 74 broadcast messages in that episode originated from the two cooperative agents, A2 and A3, who coordinated explicitly; A2 broadcasting "I'll broadcast to coordinate with A3 and establish a cooperative approach, then we can work together to efficiently collect the nearby resources." Meanwhile, the neutral agent (A1) and aggressive agent (A4) conducted a sustained mutual raiding loop for the majority of the episode, each repeatedly stealing from the other. The two cooperative agents collected resources while the other two fought, resulting in a higher efficiency of 23.3% — still well below simple, but illustrating that cooperative communication did produce some benefit within the group of agents that chose to use it.

Cooperative Vulnerability. Episode `llm_comms_seed001` illustrated the vulnerability of

cooperative agents in a mixed group. A3, the sole cooperative agent, broadcast repeated coordination attempts across multiple consecutive ticks, "I'll broadcast to coordinate with nearby agents rather than raid. Since A2 is very close with 9 resources...", but received no response. A2, the neutral agent, raided A3 repeatedly until the `raid_empty` result appeared, then continued attempting to raid anyway. A4, the deceptive agent, quietly moved around the map and collected the highest inventory of any agent in the episode (5 resources), while A1, despite also being cooperative, spent multiple consecutive ticks attempting to move into a wall. The cooperative agent's broadcasts, ignored by all, highlight a core vulnerability: communication is only beneficial if other agents are receptive to it.

6. DISCUSSION

6.1. Effect of Communication on Collective Performance

Addressing the first research question directly: enabling communication did not increase total resources collectively gathered. The communicating condition averaged lower efficiency (0.297) than the silent condition (0.397), contradicting the initial hypothesis. The second research question, whether communication shifts behaviour toward cooperation or competition, produced a nuanced answer. Communication did enable genuine cooperation between agents who shared cooperative personalities, as seen in seed010. However, across the full distribution of trials, it did not reliably reduce raiding or improve collective outcomes, and in several cases appeared to make things worse.

The scale of the gap between simple and LLM agents is the most significant finding. A four-rule greedy algorithm outperformed LLM-powered agents by roughly $2.5\times$ on average. This is consistent with observations by Guo et al., who note that LLM agents tend to underperform on tasks requiring consistent, repetitive execution [9]. Resource collection in this environment is fundamentally such a task; the near-optimal strategy involves moving toward the nearest resource and collecting it, with minimal deliberation required. The LLM's tendency to reason about personalities, model other agents' intentions, and consider communication represents overhead rather than value. This observation is in tension with the finding from Akata et al. that LLM agents tend toward cooperation when communication is available [4]. In this project, communication did not reliably produce cooperation, likely because the competitive pressures and personality diversity created conditions where deception and aggression dominated.

6.2 Emergent Behavioural Patterns

The qualitative episode analysis reveals three emergent behavioural patterns worth discussing.

Raid Loops. The first is the raid loop, observed most extremely in seed012 and `llm_silent_seed012`. When deceptive and aggressive agents are grouped together with no cooperative agents to moderate behaviour, they enter sustained mutual raiding cycles that consume nearly all available turns. No agent benefits — each raid redistributes rather than creates resources — yet agents continue raiding, apparently because the LLM's reasoning prioritises short-term gain over the globally suboptimal outcome. This behaviour has no analogue in the simple agent, which only raids when the target has strictly more inventory and stops when it does not.

Cooperative Agent Vulnerability. The second pattern is personality-driven communication segregation, seen clearly in seed010. Cooperative agents identified each other through observed behaviour and explicitly chose to coordinate, while aggressive and neutral agents ignored the communication channel entirely. This suggests personality prompts are functioning as intended — agents are genuinely behaving differently — but that the benefit of cooperative communication is limited to the cooperative subgroup. In a mixed population, cooperative agents are effectively playing a different game from their neighbours.

Inequality and Exploitation. The third pattern is the vulnerability of cooperative agents in hostile environments. Seed001 demonstrated that a cooperative agent broadcasting coordination attempts to unresponsive or actively hostile neighbours gains nothing from communication and may lose resources through the opportunity cost of broadcast turns. Cooperative communication is only beneficial when there is a receptive audience, which cannot be guaranteed with random personality assignment.

The Gini coefficient analysis adds further nuance. The higher inequality in the communicating condition compared to silent suggests that communication enabled dominant agents — particularly those with neutral or deceptive personalities who observed cooperative broadcasts without reciprocating — to exploit cooperative agents more effectively. The deceptive agent in seed001 quietly accumulated the highest inventory while cooperative agents broadcast to each other and neutral agents raided indiscriminately. This is consistent with the theoretical prediction from Panait and Luke that communication can be counterproductive in competitive settings [2] — it provides information that can be exploited as readily as it can be used for genuine coordination.

Limitations. Several limitations should be acknowledged. The use of Claude Haiku rather than a locally deployed model means the results reflect that specific model's behavioural tendencies, and larger or differently trained models might produce different outcomes. The 50-tick episode limit may be insufficient for cooperative strategies to develop trust over time — a longer episode might allow cooperative agents to establish stable alliances before aggressive agents dominate. The relatively small grid with 4 agents and 30 resources creates high

encounter density, which may have amplified raid frequency compared to a sparser environment. Finally, the LLM comms condition used only 20 trials rather than 30 due to a runtime interruption, which limits the statistical power of comparisons involving that condition.

7. CONCLUSION

7.1 Summary of Findings

This project built a multi-agent simulation in which LLM-powered agents with distinct personality profiles competed to collect resources in a grid world, with and without natural language communication. Three conditions were compared: a simple rule-based agent, an LLM agent without communication, and an LLM agent with communication enabled.

The central finding is that the simple agent substantially outperformed both LLM conditions, collecting an average of 97.6% of resources compared to 39.7% and 29.7% for the silent and communicating conditions respectively. Contrary to the initial hypothesis, enabling communication was associated with worse average performance, driven by agents consuming turns on broadcasts and by deceptive and aggressive agents exploiting the communication channel. Qualitative analysis of individual episodes revealed three emergent behavioural patterns — raid loops, personality-driven communication segregation, and cooperative agent vulnerability — that collectively explain why communication failed to improve collective outcomes in this setting.

7.2 Implications

The work suggests that the benefit of LLM communication in multi-agent systems is highly context-dependent. In tasks where the optimal strategy is simple and repetitive, LLM reasoning introduces overhead. In mixed-personality populations, communication benefits are limited to the subgroup willing to reciprocate, while potentially providing exploitable information to others.

7.3 Future Work

Future work should explore longer episodes to allow cooperative strategies more time to develop, larger and sparser environments that create genuine information asymmetry, and tasks with coordination requirements that cannot be solved by individual greedy behaviour. Controlling personality composition across trials would also allow cleaner analysis of how group composition affects communication dynamics. The most interesting open question raised by this work is not whether communication helps, but under what specific conditions it does — and the emergent behaviours observed here suggest that answer depends heavily on who is in the room.

8. REFERENCES

- [1] M. J. Mataric, "Designing emergent behaviors: From local interactions to collective intelligence," in Proc. 2nd Int. Conf. Simulation of Adaptive Behavior, 1992, pp. 275–295.
- [2] L. Panait and S. Luke, "Cooperative multi-agent learning: The state of the art," *Autonomous Agents and Multi-Agent Systems*, vol. 11, no. 3, pp. 387–434, 2005.
- [3] J. S. Park et al., "Generative agents: Interactive simulacra of human behavior," in Proc. 36th ACM Symp. User Interface Software and Technology (UIST), San Francisco, CA, USA, 2023.
- [4] E. Akata et al., "Playing repeated games with large language models," arXiv preprint, 2023. <https://arxiv.org/abs/2305.16867> (Accessed: May 21, 2026).
- [5] M. Safdari et al., "Personality as a lens: Role-playing and simulation in large language models," arXiv preprint, 2023. <https://arxiv.org/abs/2307.00184> (Accessed: May 13, 2026).
- [6] R. Axelrod, *The Evolution of Cooperation*. New York, NY, USA: Basic Books, 1984.
- [7] C. Farabet, T. Warkentin, "Introducing Gemma 3: The most capable model you can run on a single GPU or TPU", Google 2025. <https://blog.google/technology/developers/gemma-3/> (Accessed: May 14, 2026).
- [8] Anthropic, "Claude 3.5 Haiku," Anthropic.com, 2024. <https://www.anthropic.com/claude/haiku>. (Accessed: May 19, 2026).
- [9] T. Guo, X. Chen, Y. Wang, R. Chang, S. Pei, N. V. Chawla, O. Wiest, and X. Zhang., "Large language model based multi-agents: A survey of progress and challenges." arXiv preprint arXiv:2402.01680, 2024.